

BRUGGEN, Germany

WMO: 104010

Lat: 51.200N

Lon: 6.133E

Elev: 249

StdP: 14.56

Time Zone: 1.00 (EUC)

Period: 92-01

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.8	22.3	11.5	10.0	22.3	15.9	12.4	25.0	27.1	47.3	24.3	46.5	6.4	70	0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.3	86.4	67.1	82.5	65.7	78.8	64.5	69.6	81.1	67.7	78.9	66.0	75.9	7.1	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.6	95.4	74.1	63.9	89.7	71.9	62.3	84.6	70.1	33.7	80.9	32.2	79.8	30.8	76.1	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.2	18.5	16.3	DB	12.3	92.2	5.7	3.8	8.2	95.0	4.8	97.2	1.6	99.3	-2.5	102.0	(4)
(5)			WB	11.2	73.6	5.4	2.7	7.3	75.6	4.1	77.2	1.1	78.7	-2.8	80.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.8	37.5	38.9	43.5	49.2	57.6	61.1	65.3	65.4	58.4	50.4	42.7	38.2		
	DBStd	12.09	7.97	7.18	5.82	6.84	6.93	6.18	6.01	6.14	4.85	6.05	7.06	7.51		
	HDD50	1694	389	313	214	95	12	0	0	0	1	71	230	369		
	HDD65	5419	853	730	667	475	245	151	71	72	206	451	669	829		
	CDD50	1971	1	2	12	71	248	334	474	476	253	84	11	5		
	CDD65	223	0	0	0	1	17	35	79	83	8	0	0	0		
	CDH74	2092	0	0	0	18	182	359	748	736	47	2	0	0		
	CDH80	616	0	0	0	0	30	93	235	250	8	0	0	0		
(14) Wind	WSAvg	7.3	8.6	8.4	8.3	7.3	6.9	6.6	6.5	6.1	6.5	7.0	7.3	8.5		
(15) Precipitation	PrecAvg	30.0	2.5	2.3	2.0	1.7	2.3	2.8	3.0	3.1	2.3	2.4	2.6	2.9		
	PrecMax	38.3	5.4	5.4	3.7	4.1	4.4	7.4	6.4	6.4	6.3	6.1	4.3	6.2		
	PrecMin	23.5	0.2	0.4	0.1	0.0	0.7	0.6	1.3	0.7	0.5	1.3	0.2	1.0		
	PrecStd	3.7	1.2	1.2	1.0	0.9	1.0	1.5	1.4	1.7	1.5	0.9	1.0	1.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.4	57.4	64.1	77.2	84.0	88.1	91.1	91.2	79.8	70.5	60.4	56.2		
		MCWB	50.8	49.5	52.0	55.5	63.6	69.6	69.9	70.0	64.8	62.0	53.5	51.7		
	2%	DB	52.6	53.8	59.8	70.6	78.7	82.1	86.1	86.7	73.7	65.5	56.2	54.2		
		MCWB	48.7	47.7	50.1	55.9	62.0	64.8	67.7	67.2	62.9	59.1	52.8	51.0		
	5%	DB	50.2	51.3	55.8	64.9	75.1	78.1	81.9	82.2	69.8	62.0	54.3	51.8		
		MCWB	47.1	47.2	48.8	53.7	59.7	63.5	66.3	66.3	60.5	56.8	51.3	48.9		
	10%	DB	48.0	49.0	52.7	60.3	71.0	73.6	78.0	77.4	66.6	59.5	51.9	49.5		
		MCWB	45.1	45.7	47.8	51.3	58.5	61.4	65.3	64.9	59.0	55.4	49.1	46.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.9	51.0	54.0	60.5	66.5	70.0	72.7	72.1	67.1	63.1	55.5	53.5		
		MCDB	54.4	55.1	60.2	70.7	79.6	83.0	85.5	86.7	75.6	69.2	57.2	54.9		
	2%	WB	49.2	49.1	51.7	56.7	63.6	67.3	69.9	69.5	64.1	60.3	53.3	51.3		
		MCDB	51.7	52.4	56.6	66.1	75.5	79.4	81.4	80.8	71.0	64.0	55.5	53.8		
	5%	WB	47.5	47.8	50.1	54.7	61.3	64.9	68.0	67.4	62.3	58.2	51.7	49.2		
		MCDB	49.9	50.8	54.2	62.7	71.3	74.8	79.3	78.0	68.0	61.2	53.9	51.5		
	10%	WB	45.3	45.9	48.5	52.8	59.3	62.4	66.0	65.7	60.5	55.9	49.5	47.0		
		MCDB	47.7	48.7	52.4	59.3	68.6	70.6	75.7	75.0	65.5	58.8	51.3	49.4		
(35) Mean Daily Temperature Range	MDBR		9.3	10.4	13.0	15.5	17.5	17.3	17.3	17.8	14.1	12.7	9.5	8.2		
	5% DB	MCDBR	11.7	12.7	18.7	21.7	23.9	24.0	23.9	24.7	18.7	15.8	11.2	10.0		
		MCWBR	9.2	8.7	11.3	10.4	10.9	10.8	9.8	9.8	9.5	10.0	8.6	8.7		
	5% WB	MCDBR	11.4	11.2	14.5	18.7	21.7	21.0	22.3	21.1	16.5	14.4	10.3	9.6		
(40) Clear-Sky Solar Irradiance	taub		0.325	0.348	0.378	0.413	0.418	0.418	0.425	0.416	0.388	0.362	0.338	0.321		
	taud		2.403	2.365	2.285	2.213	2.238	2.255	2.236	2.285	2.353	2.417	2.404	2.372		
	Ebn at Noon		219	245	258	261	266	266	262	258	252	237	211	199		
	Edh at Noon		20	26	34	41	42	41	42	37	31	25	19	17		
(44) All-Sky Solar Radiation	RadAvg		236	451	811	1302	1561	1691	1601	1366	1007	592	278	181		
	RadStd		34	87	121	147	141	148	196	128	112	53	34	24		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (86 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page